

Q2 B: Define a helix and prove that a curve is a helix if and only if the ratio of curvature to torsion is constant.

Preliminaries: Frenet–Serret Formulas

Let $\mathbf{r}(s)$ be a smooth curve in $\mathbb{R}^3$ parameterized by arc length s , with $\kappa(s) > 0$ everywhere (so that $\mathbf{N}$ and $\mathbf{B}$ are well-defined). The Frenet–Serret frame $\{\mathbf{T}, \mathbf{N}, \mathbf{B}\}$ satisfies:

$$\mathbf{T}' = \kappa \mathbf{N}, \quad \mathbf{N}' = -\kappa \mathbf{T} + \tau \mathbf{B}, \quad \mathbf{B}' = -\tau \mathbf{N},$$

where $\kappa > 0$ is the curvature and τ is the torsion, and primes denote d/ds .

Definition of Helices

Definition 1 (Helix). *A smooth curve $\mathbf{r}(s)$ (with $\kappa > 0$) is called a **helix** (or general helix) if its unit tangent vector $\mathbf{T}(s)$ makes a constant angle α with some fixed unit vector $\mathbf{u} \in \mathbb{R}^3$, i.e.,*

$$\mathbf{T}(s) \cdot \mathbf{u} = \cos \alpha \quad \text{for all } s,$$

where $0 \leq \alpha \leq \pi$ is constant.

To Proof:

Theorem 1 (Theorem). *A smooth curve $\mathbf{r}(s)$ with $\kappa > 0$ is a helix if and only if the ratio κ/τ is constant (equivalently, τ/κ is constant).*

Proof. ($\Rightarrow$) **Assume the curve is a helix.**

Let $\mathbf{u}$ be the fixed unit vector with $\mathbf{T} \cdot \mathbf{u} = \cos \alpha$ for all s . Differentiating with respect to s :

$$\frac{d}{ds}(\mathbf{T} \cdot \mathbf{u}) = \mathbf{T}' \cdot \mathbf{u} = \kappa \mathbf{N} \cdot \mathbf{u} = 0.$$

Since $\kappa > 0$, we conclude:

$$\mathbf{N} \cdot \mathbf{u} = 0. \tag{1}$$

Now differentiate (1) with respect to s :

$$\frac{d}{ds}(\mathbf{N} \cdot \mathbf{u}) = \mathbf{N}' \cdot \mathbf{u} = (-\kappa \mathbf{T} + \tau \mathbf{B}) \cdot \mathbf{u} = 0.$$

This gives:

$$-\kappa(\mathbf{T} \cdot \mathbf{u}) + \tau(\mathbf{B} \cdot \mathbf{u}) = 0 \implies \tau(\mathbf{B} \cdot \mathbf{u}) = \kappa \cos \alpha. \tag{2}$$

Since $\mathbf{u}$ is a unit vector and $\{\mathbf{T}, \mathbf{N}, \mathbf{B}\}$ is an orthonormal basis, we can decompose:

$$\mathbf{u} = (\mathbf{T} \cdot \mathbf{u}) \mathbf{T} + (\mathbf{N} \cdot \mathbf{u}) \mathbf{N} + (\mathbf{B} \cdot \mathbf{u}) \mathbf{B} = \cos \alpha \mathbf{T} + (\mathbf{B} \cdot \mathbf{u}) \mathbf{B}.$$

Taking the squared norm $|\mathbf{u}|^2 = 1$:

$$\cos^2 \alpha + (\mathbf{B} \cdot \mathbf{u})^2 = 1 \implies \mathbf{B} \cdot \mathbf{u} = \pm \sin \alpha. \tag{3}$$

Substituting (3) into (2) (with $\sin \alpha \neq 0$, i.e., $\alpha \neq 0, \pi$):

$$\tau(\pm \sin \alpha) = \kappa \cos \alpha \implies \frac{\kappa}{\tau} = \pm \tan \alpha = \text{constant}.$$

($\Leftarrow$) **Assume $\kappa/\tau = c$ is constant.**

Define the angle α by $\tan \alpha = c$ (so α is constant), and set:

$$\mathbf{u}(s) = \cos \alpha \mathbf{T}(s) + \sin \alpha \mathbf{B}(s).$$

We claim $\mathbf{u}$ is a constant vector. Differentiating using the Frenet-Serret formulas:

$$\begin{aligned} \frac{d\mathbf{u}}{ds} &= \cos \alpha \mathbf{T}' + \sin \alpha \mathbf{B}' \\ &= \cos \alpha (\kappa \mathbf{N}) + \sin \alpha (-\tau \mathbf{N}) \\ &= (\kappa \cos \alpha - \tau \sin \alpha) \mathbf{N}. \end{aligned}$$

Since $\tan \alpha = \kappa/\tau$, we have $\kappa \cos \alpha = \tau \sin \alpha$, so:

$$\frac{d\mathbf{u}}{ds} = (\kappa \cos \alpha - \tau \sin \alpha) \mathbf{N} = 0.$$

Hence $\mathbf{u}$ is a constant unit vector (note $|\mathbf{u}|^2 = \cos^2 \alpha + \sin^2 \alpha = 1$). Moreover:

$$\mathbf{T}(s) \cdot \mathbf{u} = \cos \alpha (\mathbf{T} \cdot \mathbf{T}) + \sin \alpha (\mathbf{B} \cdot \mathbf{T}) = \cos \alpha,$$

so $\mathbf{T}$ makes the constant angle α with the fixed direction $\mathbf{u}$. Therefore, the curve is a helix. $\square$

Remarks

1. The condition $\kappa > 0$ is essential throughout; a straight line (where $\kappa = 0$) is a degenerate helix for which the theorem does not apply in this form.
2. The case $\alpha = 0$ (or π) would mean $\mathbf{T} \parallel \mathbf{u}$ for all s , i.e., the curve is a straight line. For a non-degenerate helix, $0 < \alpha < \pi/2$.
3. The axis direction is $\mathbf{u} = \cos \alpha \mathbf{T} + \sin \alpha \mathbf{B}$, which lies in the osculating plane only if $\sin \alpha = 0$ (degenerate).
4. A *circular helix* (e.g., $\mathbf{r}(t) = (a \cos t, a \sin t, bt)$) is the special case where *both* κ and τ are individually constant, not just their ratio.